

V4.2 Toy Model 4F

Conservation-Built Relational Stress Law

Rev A - Exploratory Gravity / Metric Bridge Report

Date: 2026.05.26 | Continuation after Toy Model 4E

Status: Toy Model 4F builds a source tensor from a conservation law instead of a flexible feature fit. It shows that conservation can be enforced, but conservation alone does not make the source balance the Einstein-tensor candidate. It does not derive Einstein Field Equations, physical stress-energy, Newton's constant, gravity, dark matter, dark energy, or physical 3T.

1. Purpose

Toy Model 4E showed that a flexible tensorial source can fit the Einstein-tensor candidate well, but that fit was not a physical source law. Toy Model 4F removes the flexible fitting step and builds a source from conservation first.

The narrow test is: if the source tensor is constructed to satisfy conservation, does it naturally balance the curvature tensor, or is another constitutive law still missing?

2. Critical correction before interpretation

Conservation is necessary, but not sufficient. Many tensors can be conserved. That does not mean they are the correct source for a given curvature. A valid gravity-side source must satisfy both conditions: conservation and component-by-component balance with geometry.

3. Conservation-built source law

The source is built as a static perfect-fluid-like mixed tensor:

```
T^mu_nu = diag(-rho, p, p)

rho(r) = u(r)
p(r) is not fitted to G_mu_nu.
p(r) is solved from conservation.
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The conservation condition is:

```
nabla_mu T^mu_nu = 0
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For a static radial source in the controlled Lorentzian metric, the radial conservation equation becomes:

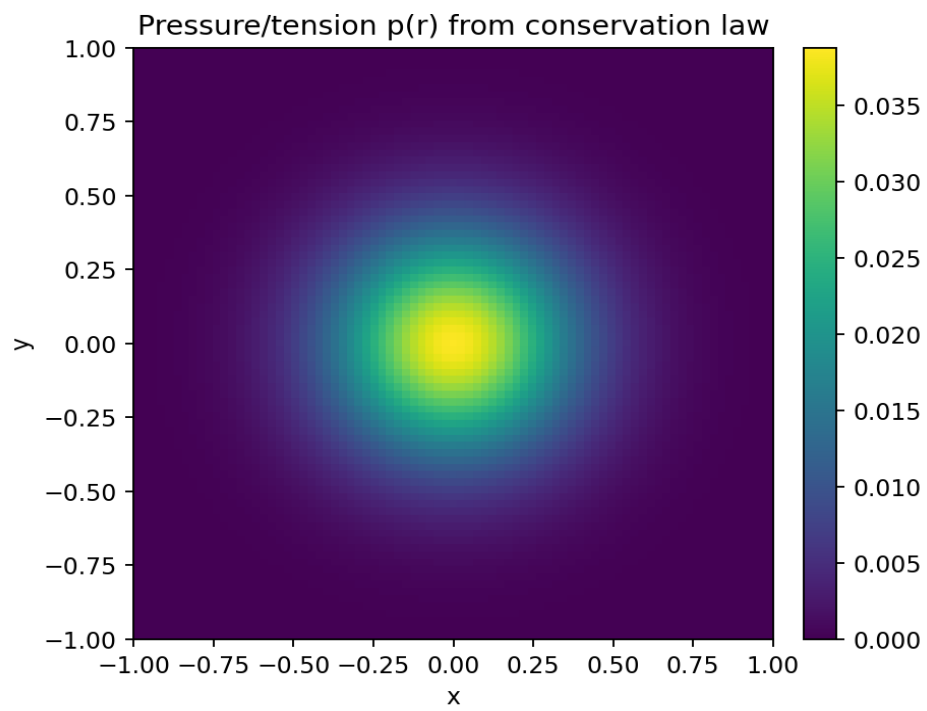
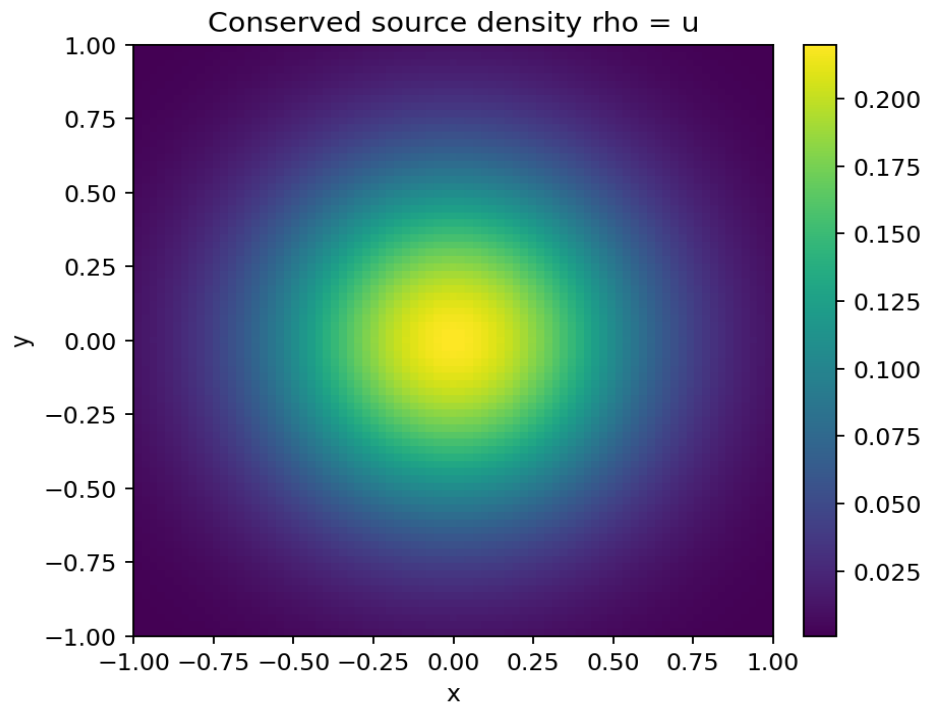
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dp/dr = -(rho + p) d[ln N]/dr

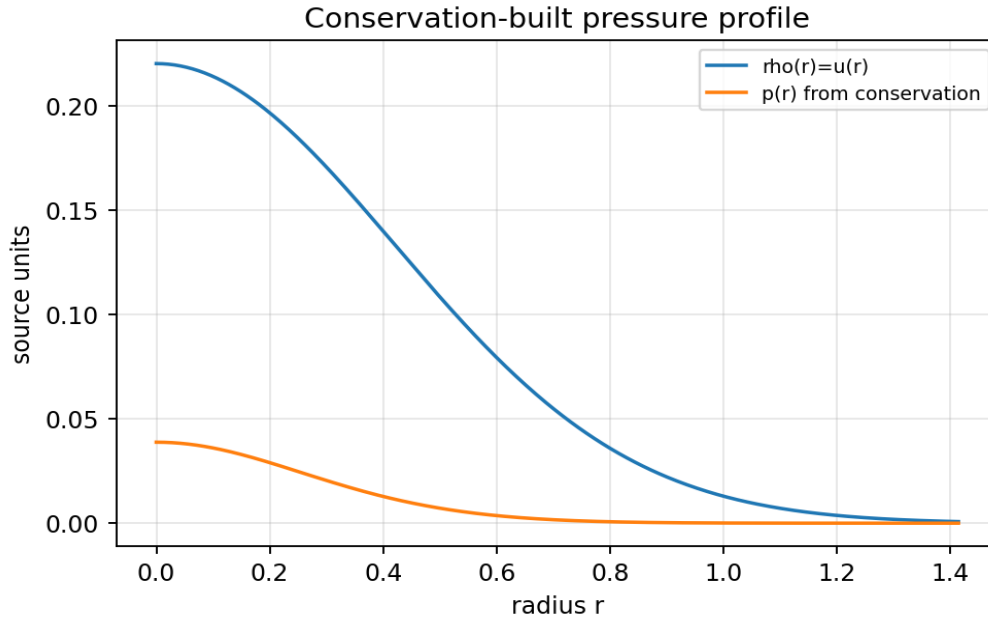
N(r) = sqrt(-g_tt) = c sqrt(1 - 2u(r))

Boundary condition:
p(Rmax) = 0
```

Meaning of symbols:

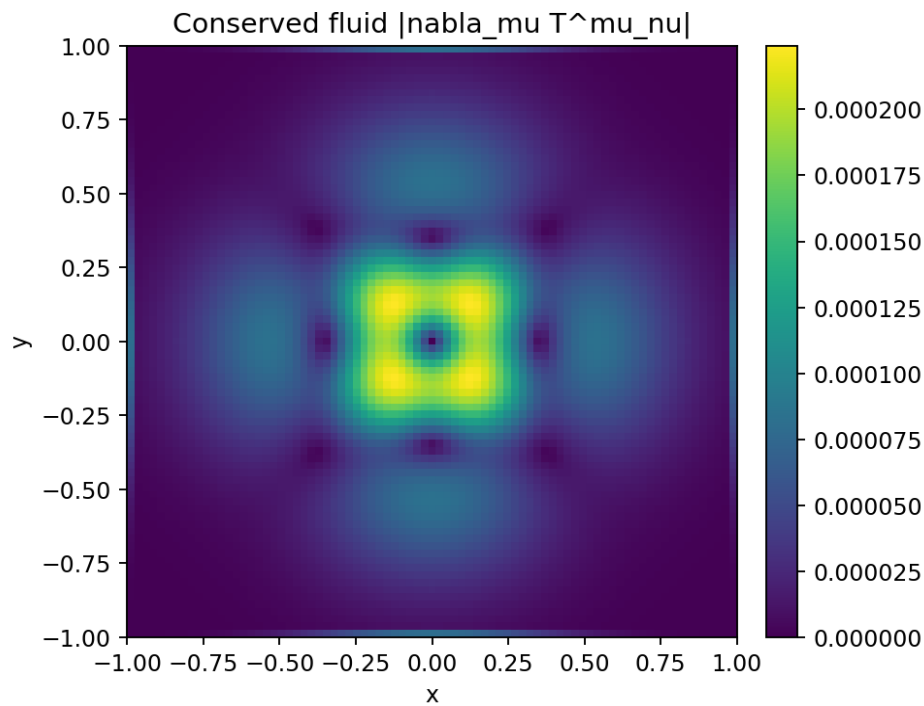
- rho is the density-like source term.
- p is pressure/tension required by conservation.
- N(r) is the lapse-like time metric factor.
- r is radial distance from the source center.
- The equation says pressure gradients balance the tendency of the density to fall through the varying time metric.





4. Conservation verification

The conserved-fluid tensor was converted into covariant and mixed forms, then tested using the same numerical covariant divergence operator used in the previous curvature reports.



Conservation diagnostic	Value
Conserved fluid max $ \nabla_\mu T^\mu_\nu $	2.241222e-04
Conserved fluid mean $ \nabla_\mu T^\mu_\nu $	4.819259e-05
Pressureless dust max divergence	7.625579e-02
4E flexible fit max divergence	1.234161e-01
Geometry-side max $ \nabla_\mu G^\mu_\nu $	1.299728e-02

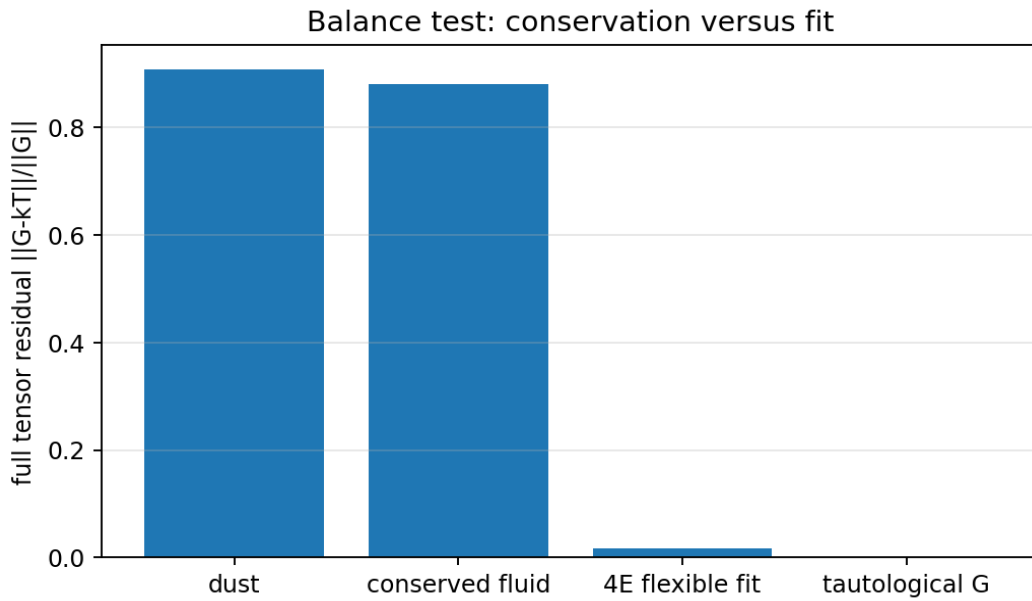
Result: the conservation-built source is much better conserved than the 4E flexible tensor fit. This confirms the construction does what it was designed to do.

5. Einstein balance test

The field-equation-like balance test is still:

$$G_{\mu\nu} \sim \kappa T_{\mu\nu}$$

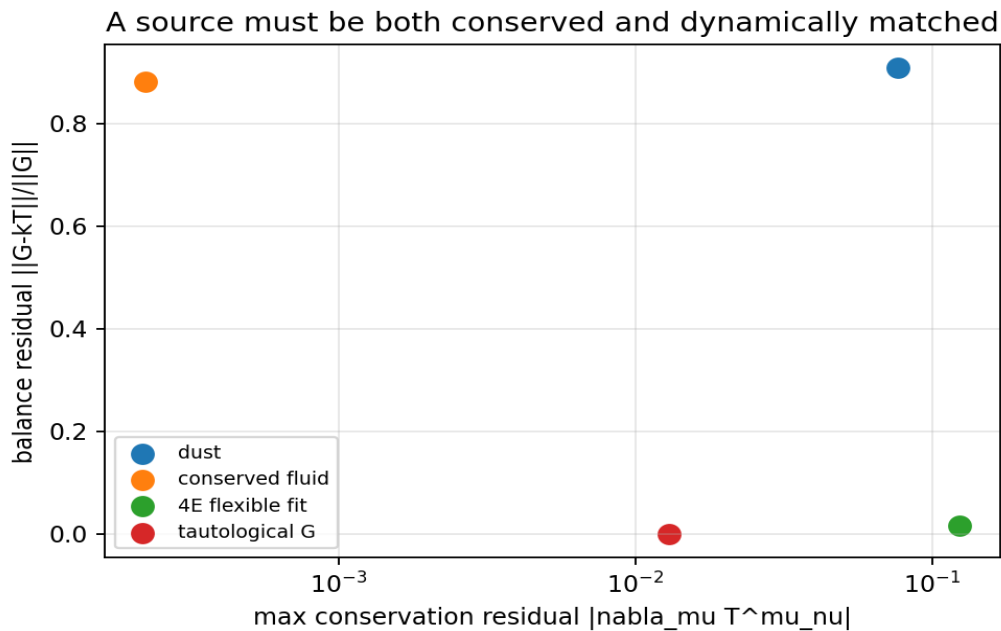
Here kappa is a fitted toy coupling constant. It is not Newton's constant and not the physical GR coupling.



Result: conservation-built fluid is conserved, but it does not balance $G_{\mu\nu}$ well. The 4E flexible fit balances much better but is not conserved well and remains a calibrated fit. The tautological G model is not an independent source law.

6. Conservation versus balance trade-off

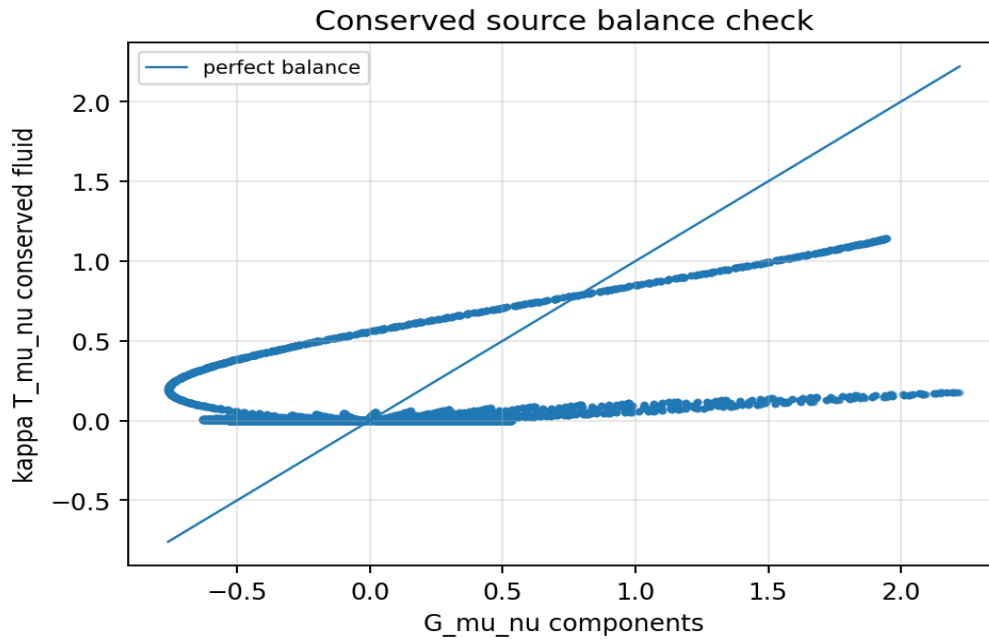
The key 4F diagnostic is the trade-off between conservation and balance.



The conserved source sits in the low-divergence/high-residual region. The flexible fit sits in the better-balance/higher-divergence region. A real source law would need to sit near both low residual and low divergence.

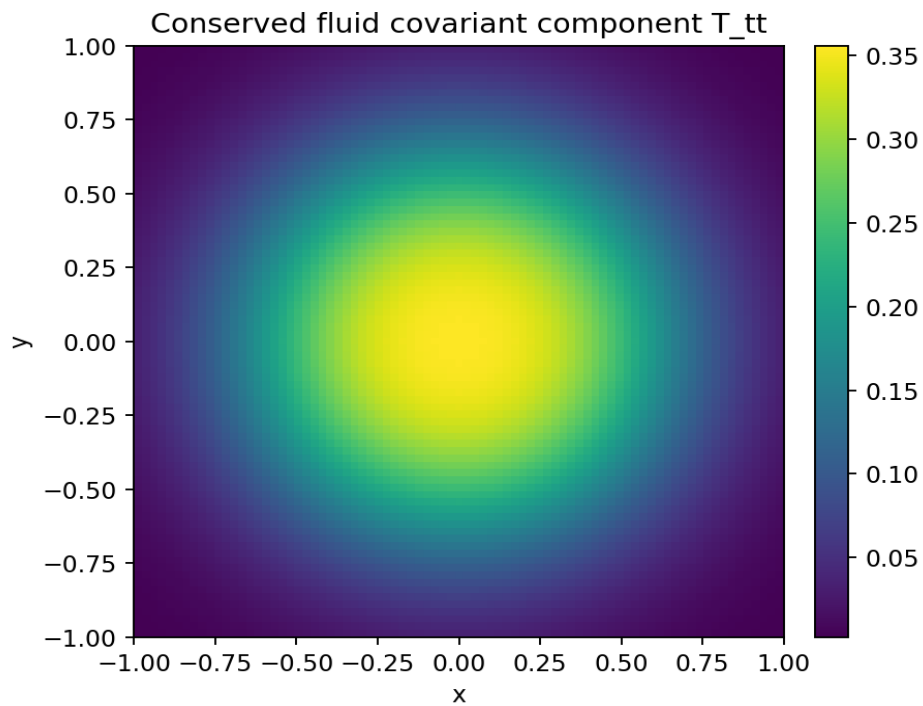
7. Component-by-component balance

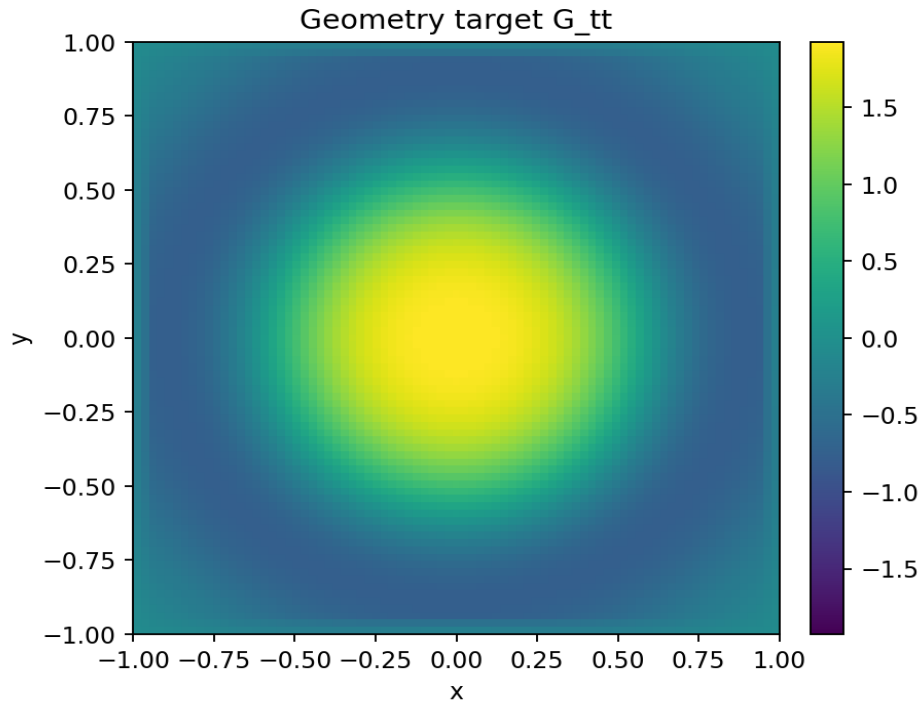
The conserved fluid was checked component by component. Its main weakness is that isotropic conserved pressure does not reproduce the full anisotropic structure of the geometry candidate.



Component	Residual	Correlation	Max G	Max kT
tt	7.946606e-01	0.972539	1.945753e+00	1.142111e+00
tx	0.000000e+00	1.000000	0.000000e+00	0.000000e+00
ty	0.000000e+00	1.000000	0.000000e+00	0.000000e+00
xx	9.374570e-01	0.881313	2.220410e+00	1.790769e-01
xy	1.000000e+00	0.000000	5.340168e-01	0.000000e+00
yy	9.374570e-01	0.881313	2.220410e+00	1.790769e-01

8. Source component maps

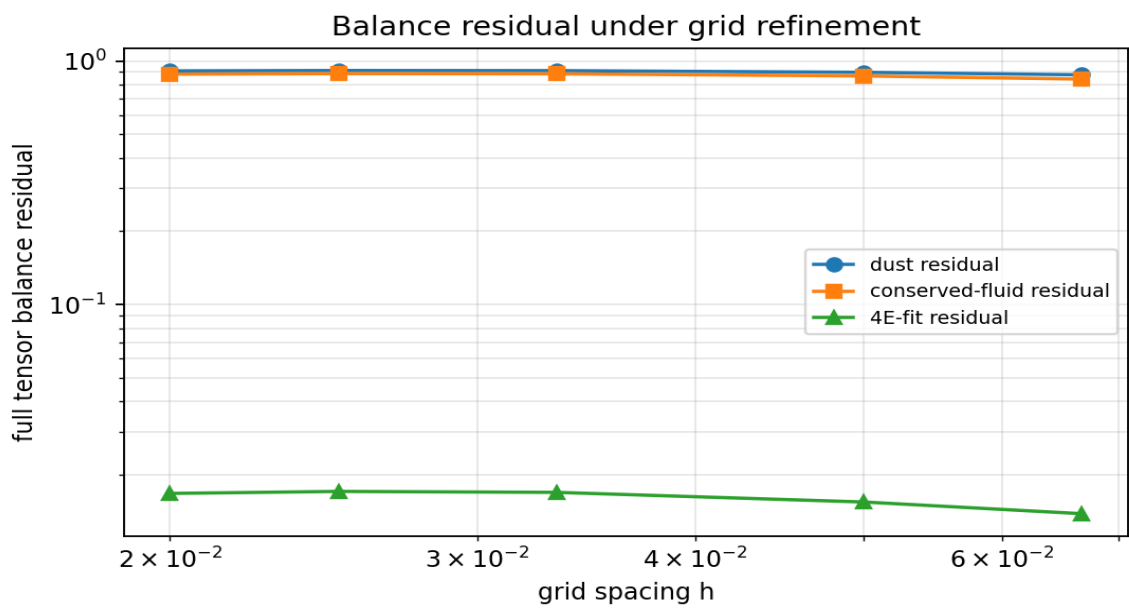




The conserved-fluid T_{tt} has the right general localization, but the full tensor pattern is still too simple to match the curvature tensor.

9. Grid-refinement behavior

The test was repeated across grid resolutions. The conclusion remains stable: the conservation-built fluid remains poorly balanced relative to the flexible fit.



N	h	dust res.	conserved res.	4E-fit res.	fluid max div	Lorentz sig.
31	0.06667	8.770780e-01	8.422288e-01	1.385039e-02	1.558333e-03	529/529
41	0.05000	8.978286e-01	8.676537e-01	1.546374e-02	8.672537e-04	1089/1089
61	0.03333	9.123284e-01	8.854550e-01	1.694247e-02	3.969000e-04	2809/2809
81	0.02500	9.134037e-01	8.867705e-01	1.709547e-02	2.241222e-04	5041/5041
101	0.02000	9.101107e-01	8.827110e-01	1.677396e-02	1.431502e-04	7569/7569

10. Core numerical record

Diagnostic	Value
Grid	81 x 81

Diagnostic	Value
Interior test points	4761
Lorentzian signatures	4761/4761
rho center	2.200000e-01
pressure center	3.876835e-02
pressure / rho at center	1.762198e-01
rho min / max	3.661316e-03 / 2.200000e-01
pressure min / max	6.454275e-06 / 3.876835e-02

11. Interpretation inside the V4.2 framework

Toy Model 4F supports this limited ladder:

```
relational density rho = u
  -> conservation-built pressure/tension p(r)
  -> source tensor T^mu_nu = diag(-rho, p, p)
  -> very small covariant divergence
  -> source is conserved

But:
conserved T_mu_nu
  -> does not naturally balance G_mu_nu in this model
```

It does not support this stronger ladder:

```
conservation alone
  -> Einstein Field Equations
  -> gravity derivation
  -> dark matter / dark energy explanation
```

12. What Toy Model 4F achieves

- It discards arbitrary source fitting and builds pressure/tension from conservation.
- It verifies that a conserved relational-fluid-like source can be constructed.
- It shows that conservation alone does not make the source balance the curvature tensor.
- It separates two requirements that were previously mixed together: conservation and geometry balance.
- It identifies the next missing object: a constitutive relational stress law linking source structure to metric response.

13. What Toy Model 4F does not achieve

- It does not derive Einstein Field Equations.
- It does not derive a physical stress-energy tensor.
- It does not derive Newton's gravitational constant.
- It does not derive the physical coupling between source and curvature.
- It does not prove gravity as relational conservation.
- It does not explain dark matter or dark energy.
- It does not prove physical 3T.

14. Close-out statement for 4F

Toy Model 4F is an important negative/diagnostic result. A conservation-built relational-fluid tensor can be constructed and it satisfies the covariant conservation test far better than the flexible 4E fit. However, it does not naturally balance the Einstein-tensor candidate. Therefore, conservation is necessary but not sufficient. The gravity bridge still lacks a constitutive relational stress law: a principled rule explaining how density, flow, pressure, tension, and shear determine the metric response. Without that law, an Einstein-equation analogue would still be fitted rather than derived.

15. Recommended next chapter

The next logical chapter is Toy Model 4G: Constitutive Relational Stress Law and Metric Response.

1. Define a source law that links relational density/flow/tension to metric components before curvature is computed.
2. Require conservation from the beginning.
3. Then compute $G_{\mu\nu}$ from the resulting metric and test balance.

4. Avoid fitting T directly to G after the fact.
5. Only if this works should an Einstein-equation analogue be discussed.

16. Rebind prompt for continuation

We are resuming after Toy Model 4F.

4E:

A flexible tensorial/differential source can fit $G_{\mu\nu}$ well, but it is a calibrated feature fit.

4F:

The flexible fit was discarded.

A conservation-built source was constructed:

$T^{\mu}_{\nu} = \text{diag}(-\rho, p, p)$

$\rho(r) = u(r)$

$p(r)$ solved from conservation:

$dp/dr = -(\rho + p) d[\ln N]/dr$

$N(r) = \sqrt{-g_{tt}}$

This made $\nabla_{\mu} T^{\mu}_{\nu}$ small.

Result:

Conservation-built source is conserved but does not balance $G_{\mu\nu}$ well.

Flexible fit balances $G_{\mu\nu}$ well but is not a principled conserved source.

Main lesson:

Conservation is necessary but not sufficient.

The missing object is a constitutive relational stress law that determines the metric response before curvature is computed.

Critical guardrail:

4F does not derive Einstein Field Equations, physical stress-energy, Newton's G, gravity, dark matter, dark energy, or physical $3T$.

Next recommended chapter:

Toy Model 4G - Constitutive Relational Stress Law and Metric Response.